

ANALYSIS2 - LIST 3: EXPANDING FUNCTIONS INTO THE TRIGONOMETRIC FOURIER SERIES

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Exercise 1. Check if those functions satisfy Dirichlet Conditions. If yes, expand them into the trigonometric Fourier series and discuss type of convergence. Notice that all of these functions are 2π periodic.

1. $f(x) = x^2, x \in \langle -\pi; \pi \rangle$

3. $f(x) = \sin ax, x \in (-\pi; \pi), a \in \mathbb{R} \setminus \mathbb{Z}$

2. $f(x) = x^3, x \in \langle -\pi; \pi \rangle$

4. $f(x) = \cos ax, x \in (-\pi; \pi), a \in \mathbb{R} \setminus \mathbb{Z}$

Exercise 2. Calculate trigonometric Fourier coefficients for the following 2π periodic functions.

1. $f(x) = \begin{cases} -x, & x \in \langle -\pi; 0 \rangle \\ x, & x \in (0; \pi) \end{cases}$

4. $f(x) = \begin{cases} -\cos x, & x \in (-\pi; 0) \\ 0, & x = 0 \\ \cos x, & x \in (0; \pi) \end{cases}$

2. $f(x) = \begin{cases} 0, & x \in (-\pi; 0) \\ x, & x \in \langle 0; \pi \rangle \end{cases}$

5. $f(x) = \begin{cases} -c, & x \in (-\pi; 0), \quad c > 0 \\ 0, & x = 0 \\ c, & x \in (0; \pi) \end{cases}$

3. $f(x) = \begin{cases} 2x, & x \in (-\pi; 0) \\ 6x, & x \in (0; \pi) \end{cases}$

6. $f(x) = \begin{cases} 0, & x \in (-\pi; 0) \\ \sin x, & x \in (0; \pi) \end{cases}$

Exercise 3. Expand $f(x) = \begin{cases} 0, & x \in (0; \frac{\pi}{2}) \\ \pi, & x \in (\frac{\pi}{2}; \pi) \end{cases}$ into the sinus Fourier series.

Exercise 4. Expand $f(x) = \begin{cases} 0, & x \in (0; \frac{\pi}{2}) \\ x, & x \in (\frac{\pi}{2}; \pi) \end{cases}$ into the cosine Fourier series.

Exercise 5. Expand $f(x) = \begin{cases} x, & x \in (0; \frac{\pi}{2}) \\ 0, & x \in (\frac{\pi}{2}; \pi) \end{cases}$ into the sinus Fourier series.

Exercise 6. Expand $f(x) = \begin{cases} x, & x \in (0; \frac{\pi}{2}) \\ 0, & x \in (\frac{\pi}{2}; \pi) \end{cases}$ into the cosine Fourier series.

Exercise 7. Expand $f(x) = x, x \in (-\pi; \pi)$ into the Fourier series. Using the expansion calculate the sums of the following number series $\sum_{n=1}^{+\infty} \frac{1}{n^2}, \sum_{n=1}^{+\infty} \frac{(-1)^n}{n^2}, \sum_{n=1}^{+\infty} \frac{1}{(2n-1)^2}$.

Exercise 8. Expand $f(x) = \frac{\pi}{4}, x \in (-\pi; \pi)$ into the Fourier series. Using the expansion calculate the sums $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$.

Exercise 9. Expand $f(x) = x^4, x \in (-\pi; \pi)$ into the Fourier series. Using the expansion calculate the sum $\sum_{n=1}^{+\infty} \frac{1}{n^4}$.